Physics-Based Fault Classification in Capillary-Driven Paper Microfluidics Using Multi-Point Electrode Sensing and Real-Time Hardware

BioMEMS Course Project Proposal | Group X | Members: _____
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1. Problem statement

Paper-based rapid diagnostic tests (RDTs) move sample and reagents by capillary wicking alone, with no pump. This makes them cheap, but failures are *silent*: too little sample, a defect in the paper or wax barrier, or a sample with abnormal viscosity can each give an invalid result that looks valid. Bezinge *et al.* [1] showed that electrodes embedded in a paper RDT can track reagent flow and flag incorrect volumes or abnormal viscosities. Practical checks, however, reduce flow to one number ("did liquid arrive before a timeout?" or a single flow time). One number cannot say *where* a problem is, nor separate causes that the physics makes distinguishable.

2. Hypothesis

Liquid-arrival times at several electrodes along a lateral paper strip, compared with a calibrated Lucas–Washburn/Darcy model, can (a) **detect** filling faults with fewer missed and false alarms than a fixed timeout or a single-sensor threshold, and (b) **classify** them into physically distinguishable classes (uniform fluid-property change; localized flow restriction *with its location*; stalled front; sensor fault), in a small fixed-point hardware block.

3. Physics basis

Washburn [2]: the wetted length obeys $L^2 = K\,t$, with $K = \gamma\, r \cos\theta / (2\mu)$. For paper, K is an *effective* Darcy coefficient fitted to published wicking data [3, 4]. The expected time to fill the segment between electrodes $i-1$ and i is $\Delta t_i = (L_i^2 - L_{i-1}^2)/K$, so a fluid-property change scales every segment equally, while a local restriction first appears in the segment that contains it. Scaling (Bond, capillary and pore-scale Reynolds numbers all far below 1) justifies the quasi-static, inertia-free model. **Identifiability limit:** arrival times reveal only K and its spatial pattern, so viscosity cannot be told apart from surface tension or contact angle, and a restriction upstream of the first electrode mimics a fluid change. These limits will be stated explicitly.

4. Objectives

1. Build a capillary-filling model of a paper strip, calibrated to **published** wicking data, with a sourced scaling analysis.
2. Derive an **identifiability table**: which faults multi-point arrival times can and cannot distinguish.
3. Design a model-based detector/classifier and two baselines (fixed timeout; single-sensor threshold).
4. Implement the detector as **synthesizable SystemVerilog RTL**, verified bit-exactly against an independent Python golden model, with assertions and coverage.
5. Quantify detection, classification and localization under noise, and the hardware cost (FPGA resources, timing).

5. Methodology

Python physics model (Washburn + Darcy series resistance, finite sample volume, fluid-property changes, strip-to-strip variability, sensor jitter and failures; richer than the detector, so validation is not circular) → **seeded fault-scenario corpus** (each class × location × noise level) → **golden detector** (floating and fixed point) and baselines → **RTL**: timestamp unit, per-segment ratio test by cross-multiplication (no divider), online estimate of K, flow-validation FSM with model-derived adaptive timeouts, rule-based classifier → **verification** in Vivado xsim (file-driven corpus regression, SVA, functional coverage) → **synthesis** for Zynq-7020 (PYNQ-Z2). Thresholds are tuned on one seed set and evaluated on a held-out set. Real electrode traces released with [1] will be used to check the event-detection front end.

6. Expected outcomes and deliverables

Physics	Calibrated L^2–t fit with sourced K; scaling table; identifiability table
Performance	Detection/false-alarm rate vs. noise for all 3 methods; confusion matrix; localization accuracy; latency
Hardware	RTL bit-exact with the golden model on 100% of scenarios; LUT/FF/DSP and Fmax report; optional PYNQ-Z2 replay demo

7. Relevance and novelty

Applies core course physics (capillarity, Darcy flow in porous media, dimensionless scaling) to an active research area: paper and electrochemical RDTs (course topics #3, #8, #12; sensor-data processing #44). It extends published flow monitoring [1] from single flow times to **multi-point fault localization with an explicit identifiability analysis**, implemented as real-time hardware.

8. Plan

(i) Physics model, literature calibration and scaling analysis → (ii) scenario corpus, golden classifier and baselines with metrics → (iii) RTL, corpus regression, assertions, coverage and synthesis → (iv) real-trace check against [1] and report. Hardware demo on PYNQ-Z2 only if time permits. A revised 1–2 page plan follows the TA discussion (7 Nov).

[1] L. Bezinge, A. J. deMello, C.-J. Shih, D. A. Richards, "Quantitative reagent monitoring in paper-based electrochemical rapid diagnostic tests," *Lab Chip*, 24(15), 3651–3657, 2024. doi:10.1039/D4LC00390J
[2] E. W. Washburn, "The dynamics of capillary flow," *Phys. Rev.*, 17(3), 273–283, 1921.
[3] S. Mendez *et al.*, "Imbibition in porous membranes of complex shape: quasi-stationary flow in thin rectangular segments," *Langmuir*, 26(2), 1380–1385, 2010.
[4] E. Fu, S. Ramsey, P. Kauffman, B. Lutz, P. Yager, "Transport in two-dimensional paper networks," *Microfluid. Nanofluid.*, 10, 29–35, 2011.